Same Model, New Patient: Rare Arrhythmias at the Level of the Best Published Results, and the Tenfold Gap Patient Overlap Hides

Rare-Arrhythmia Detection with Generative ECG Augmentation

Course project report, 10-page version

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Abstract

Heartbeat classifiers are compared on the MIT-BIH Arrhythmia Database, and they fail where it matters most: on the rare supraventricular (S) and fusion (F) beats of patients they have never seen. We built a complete, pre-registered pipeline and report two findings. First, on 22 test patients kept out of training, our best pipeline reaches a rare-class F1 of 0.587 (S F1 0.647, F F1 0.527). That is above de Chazal et al. and Huang et al. on S and V beats, and above Zhu et al. (2022), the most recent inter-patient method we compare with, on fusion beats and on the mean of S and F. It gets there by reading each beat in the context of its own patient (the beat relative to the patient's own beats, prematurity against the patient's own rhythm, no S calls during atrial fibrillation) and by averaging ten runs. Second, we measured what testing on already-seen patients hides: the same network scores 0.837 when the test patients were also in training and 0.088 when they were not. Our pre-registered question, whether synthetic beats from a conditional VAE or a WGAN-GP help, has a clear answer: not detectably (best gain +0.025, 95% CI −0.018 to +0.047), on par with simple class balancing. Every prediction comes with a saliency map and a note whose every claim is checked against the model's output.

1 Introduction

On patients it has never seen, a heartbeat classifier misses most of the beats that matter. Trained on the MIT-BIH Arrhythmia Database [11] and tested on other patients, the classic methods find supraventricular ectopic (S) beats with an F1 of 0.511 [4] and 0.577 [8]; the most recent method we compare with reaches 0.686 for S beats and 0.477 for fusion (F) beats [16]. Our own first system, built and tested under rules registered in advance, did worse: a rare-class F1, the mean of S and F, of 0.113. This report shows how we took it to 0.587 (Section 4.1), above the classic methods on S and V beats and above the most recent one on fusion beats and on the mean of S and F.

The rare beats are hard because patients differ: a beat's shape depends on the heart, the electrode position and the disease, and S beats, whose burden goes with later atrial fibrillation and stroke [2], and F beats are both rare and concentrated in a few people. A model that has seen a few hundred S beats from a dozen people has to recognise them in someone new.

Generative models offer an apparent remedy: learn what the rare beats look like and manufacture more of them. Conditional VAEs [14] and Wasserstein GANs [6] draw convincing synthetic ECG. We registered the question before any test data was read:

With training and test patients strictly apart, does class-conditional generative augmentation improve detection of the rare classes S and F, compared with no augmentation, a class-weighted loss, random oversampling and SMOTE?

We built the whole pipeline (data, two classifiers, two generator families, augmentation, saliency maps and fact-checked notes), ran it in phases that each ended with a gate at which a rule registered in advance chose on validation patients only, and read the test patients once. Twenty-three follow-up analyses, each registered before it ran, then asked what does help; 761 runs are logged. The findings, in the order of this report:

- **Above earlier inter-patient results on the rare classes.** Our best pipeline reaches a rare-class F1 of 0.587: above de Chazal et al. and Huang et al. on S and V beats, and above Zhu et al. on fusion beats and the mean of S and F (Section 4.1).
- **Patient overlap inflates the score almost tenfold**, measured on one pipeline and three random splits (Section 4.2).
- **Synthetic beats alone do not help on new patients**: the pre-registered answer is no (Section 4.3), and synthetic beats that look more realistic are not more useful (Section 4.4).

2 Data and rules

Data. The MIT-BIH Arrhythmia Database [11, 5] holds 48 half-hour two-lead recordings whose beats were labelled by cardiologists. We use lead MLII at 360 Hz, map the labels to the AAMI classes [1] N (normal), S (supraventricular ectopic), V (ventricular ectopic) and F (fusion), and leave out the four paced records. Figure 1 shows what the classes look like.

Patients kept apart. We use the split of de Chazal et al. [4]: 22 records (DS1) for training and validation, the other 22 (DS2) for the test, so no patient is in two splits; five DS1 patients form the validation set (Table 1). Generators, oversamplers and normalisation see training patients only.

Split	Records	N	S	V	F
training	17	35,496	644	3,030	388
validation	5	10,370	300	758	27
test (DS2)	22	44,259	1,837	3,221	388

Table 1: Records and beats per split. S beats are under 4% of each split, and F beats at most 1%.

Measure and rules. The primary measure is the rare-class macro-F1, the mean F1 of S and F on the test patients, with 95% confidence intervals from a bootstrap over patients and Holm's correction [7] for the four confirmatory comparisons. Every selection rule and hypothesis was registered before the experiment it governs, and each phase ended with a gate at which the rule was applied and approved. The test file opens only for the final evaluation, and every read is logged. The follow-up analyses were registered as new versions of the rules (21 in all) before they ran, and are reported apart from the primary results.

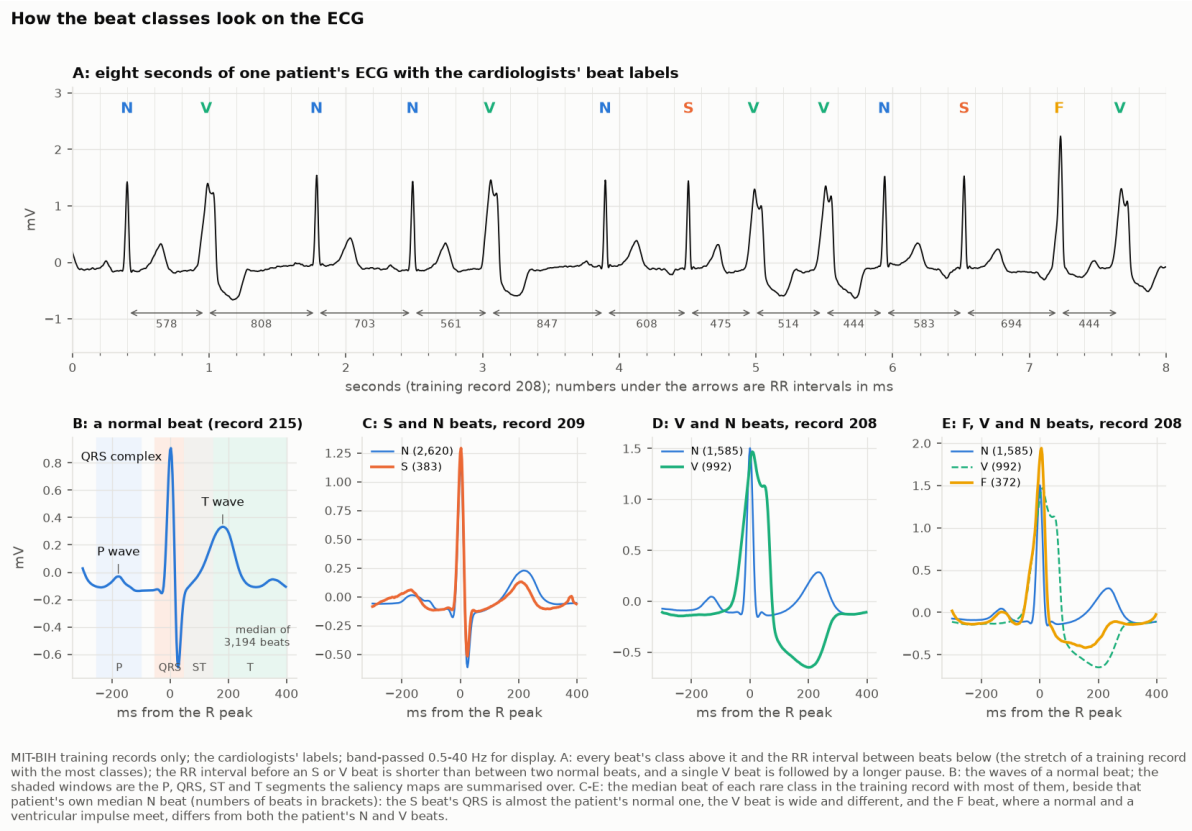


Figure 1: The beat classes on the ECG (training records only). An S beat comes early but looks almost like the patient's normal beat; a V beat is wide and different; an F beat lies between the two. The RR intervals (arrows) are what reveals an early beat.

3 The system

Figure 2 shows the pipeline: the pre-registered system on the left, and in the middle the follow-up pipeline behind our best result. Each stage was chosen on validation patients:

- **Beat:** a 0.5 s window around the R peak, z-scored, with the RR intervals before and after the beat as extra input (the best of four representations); in the best pipeline, the beat minus the patient's own template beat, and RR intervals divided by the recording's mean RR [10].
- **Classifiers:** a shallow CNN (73k parameters) and a BiLSTM (135k), the best of four architectures; pretraining on PTB-XL did not help.
- **Generators and augmentation:** a conditional VAE and a WGAN-GP, each the best of four designs, trained with three seeds into frozen banks of synthetic beats; each rare class brought up to a fraction (0.1, 0.25 or 0.5) of the N count with synthetic beats screened by dynamic time warping. With no augmentation, the class-weighted loss, random oversampling and SMOTE [3] this made a 96-run grid.
- **Decision (best pipeline):** ten independently trained runs averaged; Huang et al.'s rule calls a beat S when its RR interval is short against the recording's mean [8]; and no S call during atrial fibrillation, detected from RR intervals by a detector fitted on the MIT-BIH Atrial Fibrillation Database [5].

- **Explanation and note:** Grad-CAM [13] for the CNN and occlusion for the BiLSTM, summarised as the share of attention in the P, QRS, ST and T windows; a language model (Gemini) writes a note from a JSON of the model's output only, and a fact checker rejects any number, class or finding not in that JSON.

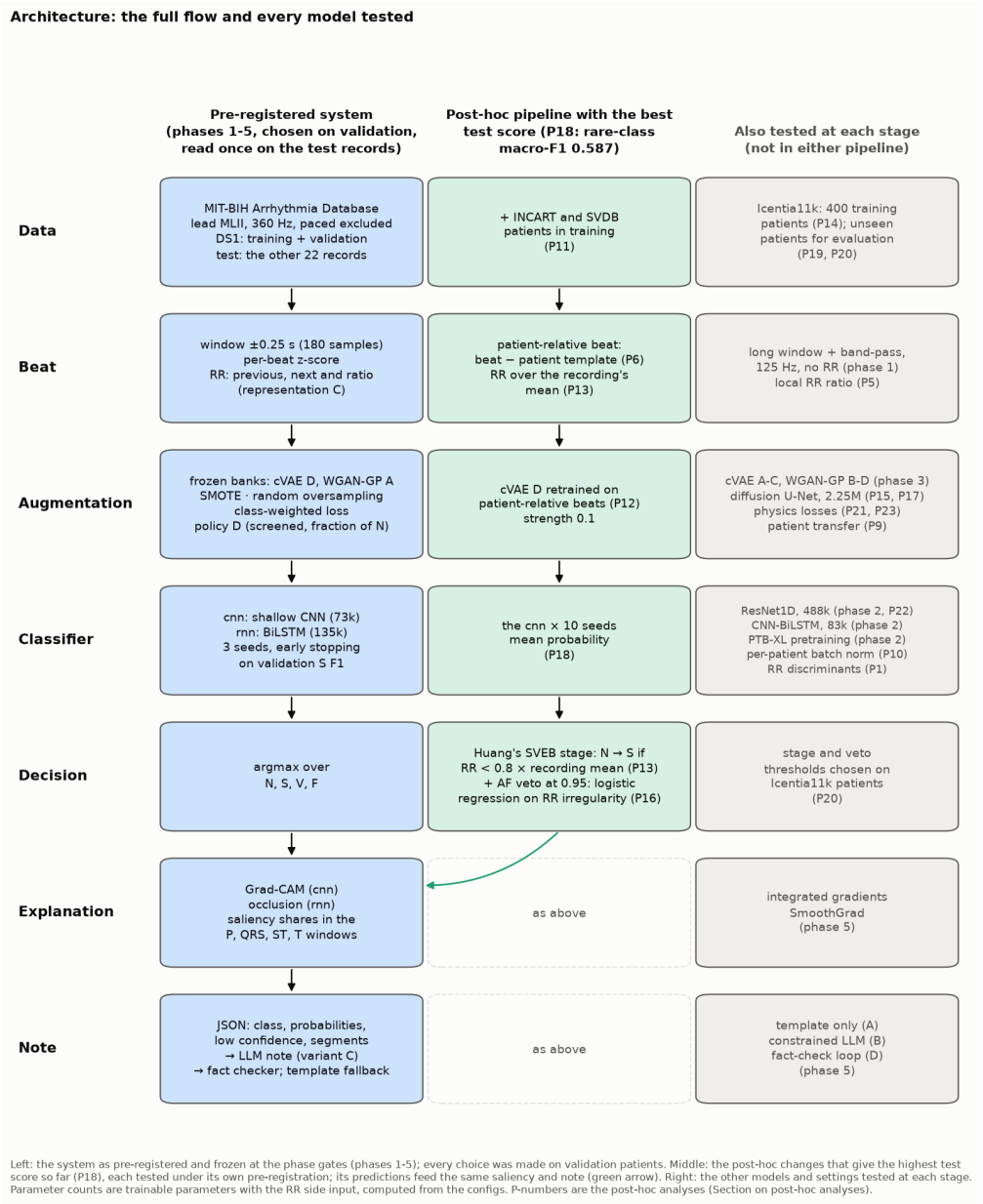


Figure 2: The pipeline. Left: the pre-registered system. Middle: the follow-up pipeline with the best test score (Section 4.1). Right: what else was tested at each stage.

4 Results

4.1 Above earlier inter-patient results on the rare classes

Twenty-three follow-up analyses (P1–P23), each registered before it ran, asked what moves the new-patient score. Four changes did, each building on the last (Figure 3, left):

1. **Each beat relative to the patient** (P7): subtracting the patient's own template beat raised the CNN without augmentation from 0.088 to 0.371.
2. **Prematurity against the patient's rhythm** (P13): RR intervals divided by the recording's mean RR, with Huang et al.'s rule on top, raised the cVAE condition to 0.523.
3. **No S calls during atrial fibrillation** (P16): the veto raised every condition by 0.018 to 0.034, each 95% interval above zero.
4. **Ten runs averaged** (P18): the cVAE's ten-run ensemble reached 0.587 [0.086, 0.702], with S F1 0.647 and F F1 0.527.

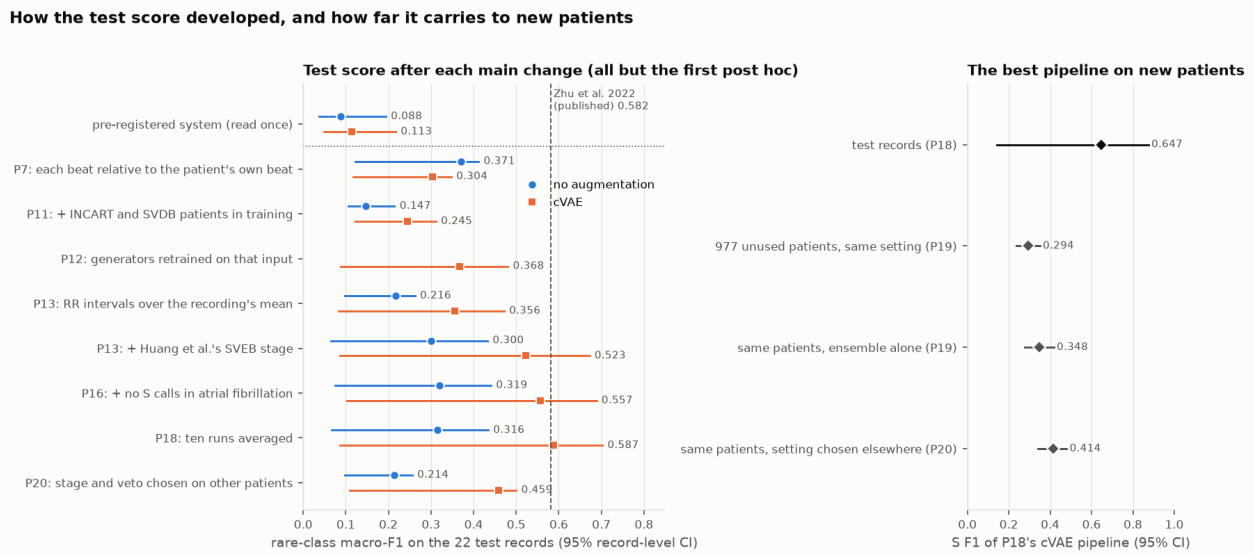


Figure 3: Left: the test score after each main change, without augmentation (circles) and with the cVAE (squares); every row below the dotted line is a follow-up analysis; dashed, the published result of Zhu et al. Right: the best pipeline's S F1 on the test patients and on unseen Icentia11k patients.

Table 2 places the best pipeline among published new-patient results. It is above the two classic methods on both classes we compare: S F1 0.647 against 0.511 and 0.577, V F1 0.936 against 0.797 and 0.924. Against Zhu et al., the most recent inter-patient method we compare with, it is higher on fusion beats (0.527 against 0.477) and on the mean of S and F (0.587 against 0.582), and lower on S (0.647 against 0.686) and V (0.936 against 0.960). Two cautions keep this in proportion: the follow-up analyses read the same 22 test patients more than once, so the score is optimistic, and its interval is wide. On 977 Icentia11k patients [15] that no analysis had used, the same pipeline reached S F1 0.294 [0.237, 0.350], and 0.414 when its two thresholds were chosen on 388 other Icentia11k patients.

	S F1	V F1	F F1	Mean of S and F F1
New patients (inter-patient, trained on DS1, tested on DS2)				
de Chazal et al. 2004 [4]	0.511	0.797	–	–
Huang et al. 2014 [8]	0.577	0.924	–	–
Zhu et al. 2022 [16]	0.686	0.960	0.477	0.582
Mousavi et al. 2019† [12]	0.907	0.997	–	–
Ours, best pipeline (P18)	0.647	0.936	0.527	0.587
Ours, pre-registered system (CNN with the cVAE)	0.173	0.893	0.054	0.113
Patients seen in training (beats split at random)				
Kwak and Jung 2024 [9]	0.947	0.985	0.903	0.925
Ours, the same CNN (no augmentation)				0.837

Table 2: Our results beside published ones (F1 computed from the reported sensitivity and positive predictivity where a paper gives no F1). †Far above every other inter-patient report; its protocol could not be checked here.

4.2 Patient overlap inflates the score almost tenfold

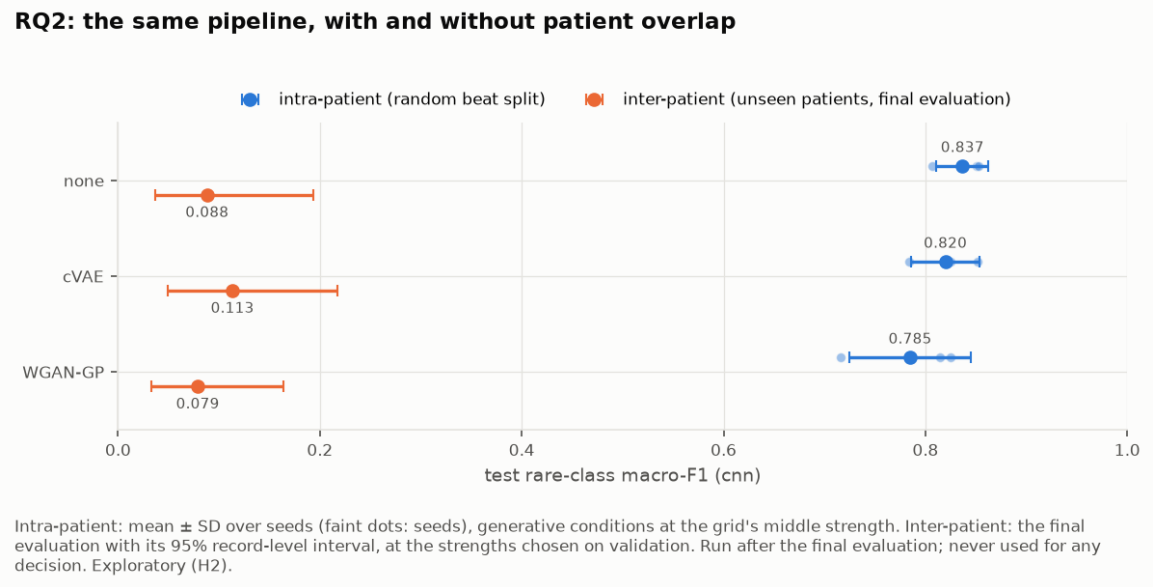


Figure 4: The same pipeline, scored on held-out beats of the training patients (blue) and on patients it has never seen (orange). The rare-class macro-F1 falls from about 0.8 to about 0.1, with or without generative augmentation.

After the final evaluation, the same CNN was trained and tested on a random beat split of all 44 records. Without augmentation it reached 0.837 ± 0.026 (Figure 4), and 0.803 and 0.777 on two further random splits, against 0.088 on new patients. Patient overlap has been known to inflate scores since de Chazal et al. introduced the inter-patient split; our measurement shows how much on one pipeline, and that generative augmentation did not help under either protocol (with patients seen in training: cVAE −0.017, WGAN-GP −0.052). It also explains Table 2: random-split results above 0.9, such as Kwak and Jung's [9], sit next to our own 0.837 with patients seen in training, not next to new-patient results.

4.3 The pre-registered answer: synthetic beats alone do not help

Four hypotheses were registered before any test record was read (Table 3), and the verdicts apply the registered rules as written. Table 4 gives the single test read. For the CNN, no augmentation scored 0.088, the cVAE 0.113 and the WGAN-GP 0.079, against 0.113 for the class-weighted loss and 0.136 for random oversampling. The cVAE's gain, +0.025 (95% CI −0.018 to +0.047), points the right way, and it nearly tripled the CNN's sensitivity for F beats (0.064 to 0.185), but no confirmatory comparison survived Holm's correction and the cVAE was level with the cheap methods. The hypothesis is not supported at this sample size.

Hypothesis	Statement	Verdict	Evidence
H1	confirmatory: a generative condition improves test rare-class macro-F1 over none (Holm, 4 tests)	not supported	no confirmatory comparison significant
H1b	exploratory: gains larger for F than for S	mixed	F gain larger than S gain in 2 of 4 generative conditions
H1c	secondary: generative beats SMOTE, random oversampling and class-weighted loss	not supported	0 of 6 comparisons of the best generative condition with a baseline significant
H2	exploratory: intra-patient evaluation inflates the baseline and the gains	not consistent	inflation of the baseline +0.748; larger intra-patient gain for 0 of 2 generators
H3	exploratory: a quality metric predicts the downstream gain	not supported	0 of 5 metrics with an interval excluding zero
H4	exploratory: augmented models are at least as faithful as the baseline	mixed	augmented faithfulness at least the baseline's in 1 of 4 model pairs

Table 3: The pre-registered hypotheses, their verdicts and the evidence, all on the 22 test patients.

Architecture	Condition	Strength	Rare macro-F1 [95% CI]	N recall
cnn	none	–	0.088 [0.037, 0.194]	0.962
cnn	class weighted	0.5	0.113 [0.057, 0.207]	0.933
cnn	random oversampling	0.1	0.136 [0.061, 0.247]	0.844
cnn	smote	0.1	0.070 [0.025, 0.172]	0.985
cnn	cvae	0.25	0.113 [0.049, 0.217]	0.938
cnn	wgan gp	0.1	0.079 [0.033, 0.164]	0.955
rnn	none	–	0.075 [0.014, 0.232]	0.961
rnn	class weighted	0.75	0.061 [0.022, 0.168]	0.865
rnn	random oversampling	0.1	0.090 [0.028, 0.196]	0.883
rnn	smote	0.1	0.086 [0.025, 0.206]	0.894
rnn	cvae	0.1	0.073 [0.035, 0.157]	0.926
rnn	wgan gp	0.5	0.069 [0.027, 0.156]	0.857

Table 4: The 22 test patients, read once: rare-class macro-F1 with 95% intervals from a bootstrap over patients, and the recall of normal beats. The strength is the fraction of the N count chosen on validation.

4.4 Why: realistic synthetic beats copy the training patients

Five quality metrics (fidelity, diversity, memorization and a classifier two-sample test), first checked on deliberately degraded beats, were compared with the downstream gain over 96 checkpoints of 24 generator runs; none predicted it (Spearman ρ from -0.20 to $+0.18$, every interval including zero). The reason shows in the beats: 57–65% of the synthetic S beats lie nearest to a single training patient, record 209, just as 60% of the real training S beats do, while the validation patients' S beats lie nearest to training patients who contribute only 1–2% of them. The generators reproduce the training patients faithfully; new patients need what the training set lacks, which is why the patient-relative changes of Section 4.1 helped where the synthetic beats did not.

4.5 A prediction a clinician can check

Of four saliency methods, the rule chose Grad-CAM for the CNN, the most faithful (deleting its most salient samples hurt the model most), and occlusion for the BiLSTM. Of four note generators tried on 50 beats, all made zero unsupported claims; the rule chose Gemini with worked examples and a fixed output schema, which covered every required fact in 2.6 s per note. The demo (Figure 5) shows the whole chain for one beat: the offline analysis by our models, then a review written by Gemini from those results only, with every claim it makes checked against them.

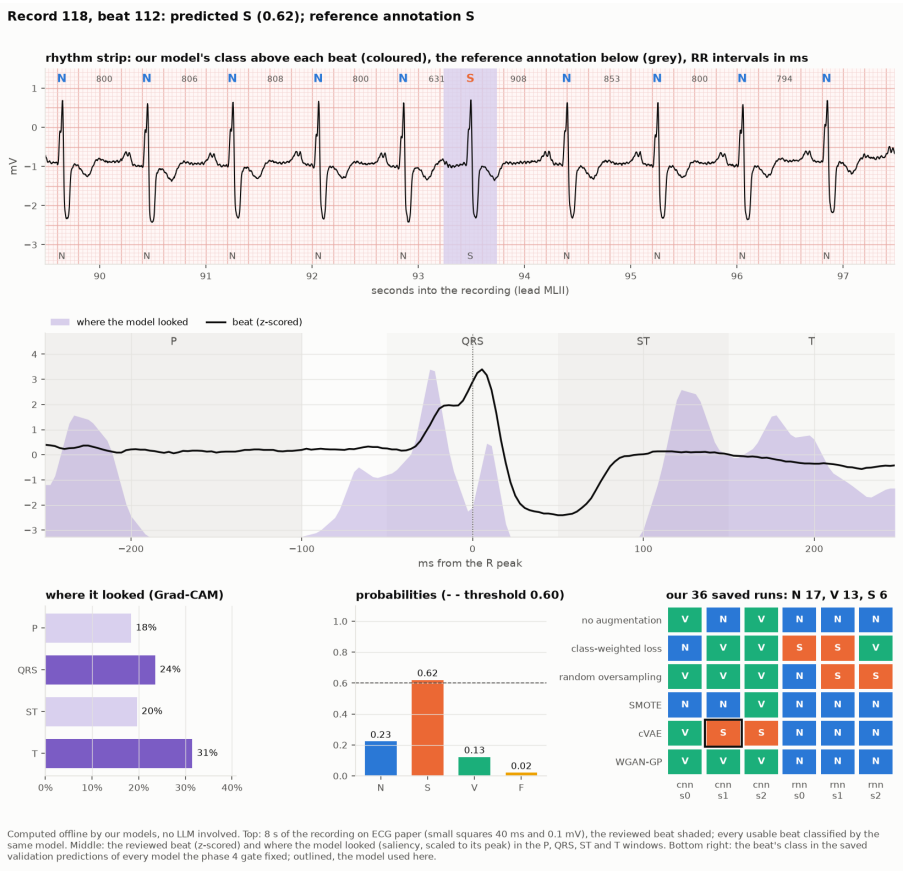


Figure 5: The demo for one S beat of a validation patient: the rhythm strip with each beat's predicted class and the reference label, where the model looked, the probabilities, and how all 36 saved runs classified this beat (only 6 called it S).

5 What it means

Context, not volume. With 17 training patients, the bottleneck for new patients is not the number of rare beats but how many different patients they come from. The generators multiplied the beats of the few patients they saw; what lifted our results above earlier inter-patient methods instead read each beat in the context of its own patient, as a cardiologist reads a rhythm strip: is this beat early for this heart, does it differ from this patient's normal beat, is the rhythm fibrillating?

The protocol decides the score. The same network scores 0.837 or 0.088 depending only on whether the test patients were seen in training. Results on random beat splits should not be compared with results on new patients.

Limitations. The rare beats sit in very few patients (most test S beats in one record), so the intervals are wide; records 201 and 202 come from the same person (the conclusions hold without record 202). The best pipeline was built after the test patients had been read once, so its test score is optimistic; the unseen Icentia11k patients give the fairer estimate. The notes restate the model; they do not diagnose.

Conclusion. Reading each beat in the context of its own patient, and averaging runs, brought rare-arrhythmia detection on new patients above the classic inter-patient methods and level with the most recent one we compare with, ahead of it on fusion beats. Synthetic heartbeats from a cVAE or a WGAN-GP did not detectably add to simple class balancing, and testing on already-seen patients would have hidden both results behind a score almost ten times higher. Whether generative augmentation can add more needs what it cannot manufacture: more patients with rare beats.

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